

Hands-on Lab: String Patterns, Sorting and Grouping in MySQL

Estimated time needed: 30 minutes

In this lab, you will learn how to create tables and load data in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Objectives

After completing this lab, you will be able to:

- Filter the output of a SELECT query by using string patterns, ranges, or sets of values.
- Sort the result set in either ascending or descending order in accordance with a pre-determined column.
- Group the outcomes of a query based on a selected parameter to further refine the response.

Software Used in this Lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



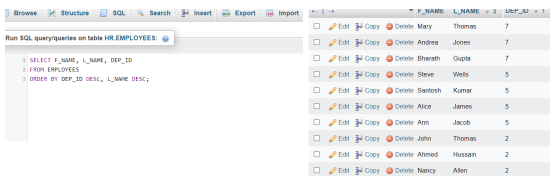
To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Database Used in this Lab

The database used in this lab is an internal database. You will be working on a sample HR database. This HR database schema consists of 5 tables called **EMPLOYEES**, **JOB_HISTORY**, **JOB**, **DEPARTMENTS** and **LOCATIONS**. Each table has a few rows of sample data. The following diagram shows the tables for the HR database:

SAMPLE HR DATABASE TABLES

EMPLOYEES						
EMP_ID	NAME	JOB_ID	START_DATE	END_DATE	MANAGER_ID	DEPT_ID
100	John	MANAGER	2001-09-17		110	10
101	James	MANAGER	2001-09-17		110	10
102	Steven	MANAGER	2001-09-17		110	10
103	Alex	MANAGER	2001-09-17		110	10
104	Samuel	MANAGER	2001-09-17		110	10
105	Michael	MANAGER	2001-09-17		110	10
106	Pat	MANAGER	2001-09-17		110	10
107	Scott	MANAGER	2001-09-17		110	10
108	Tyrael	MANAGER	2001-09-17		110	10
109	Dora	MANAGER	2001-09-17		110	10
110	Deena	MANAGER	2001-09-17		110	10
111	Anthony	MANAGER	2001-09-17		110	10
112	Jeffrey	MANAGER	2001-09-17		110	10
113	Timothy	MANAGER	2001-09-17		110	10
114	Ernest	MANAGER	2001-09-17		110	10
115	Adam	MANAGER	2001-09-17		110	10
116	Jessie	MANAGER	2001-09-17		110	10
117	Gordon	MANAGER	2001-09-17		110	10
118	Serge	MANAGER	2001-09-17		110	10
119	Shelley	MANAGER	2001-09-17		110	10
120	Walter	MANAGER	2001-09-17		110	10
121	Kenneth	MANAGER	2001-09-17		110	10
122	Howard	MANAGER	2001-09-17		110	10
123	Van	MANAGER	2001-09-17		110	10
124	Patricia	MANAGER	2001-09-17		110	10
125	Shirley	MANAGER	2001-09-17		110	10
126	Greg	MANAGER	2001-09-17		110	10
127	Cheryl	MANAGER	2001-09-17		110	10
128	David	MANAGER	2001-09-17		110	10
129	Julia	MANAGER	2001-09-17		110	10
130	Barbara	MANAGER	2001-09-17		110	10
131	Sigal	MANAGER	2001-09-17		110	10
132	Isabel	MANAGER	2001-09-17		110	10
133	Heather	MANAGER	2001-09-17		110	10
134	Doris	MANAGER	2001-09-17		110	10
135	Fred	MANAGER	2001-09-17		110	10
136	Maria	MANAGER	2001-09-17		110	10
137	Sutton	MANAGER	2001-09-17		110	10
138	Griffin	MANAGER	2001-09-17		110	10
139	Timothy	MANAGER	2001-09-17		110	10
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273	Griffin	MANAGER	2001-09-17		110	10
274	Timothy	MANAGER	2001-09-17		110	10
275	Jeffrey	MANAGER	2001-09-17		110	10
276	Ernest	MANAGER				



EMP_ID	F_NAME	L_NAME	DEP_ID
1	John	Deere	1
2	James	Clark	2
3	Michael	Burns	3
4	Wendell	Donato	4
5	Pat	Haas	5
6	Scott	Tobias	6
7	David	Turner	7
8	John	Abel	8
9	Paul	Christiansen	9
10	Jeffrey	Goldberg	10
11	Timothy	Gietz	11
12	Luis	Popp	12
13	Denise	Raisz	13
14	Neena	Koch	14

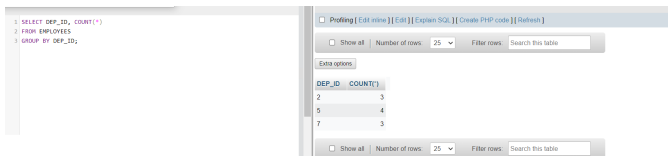
Grouping

In this exercise, you will go through some SQL problems on Grouping.

NOTE: The SQL problems in this exercise involve usage of SQL Aggregate functions AVG and COUNT. COUNT has been covered earlier. AVG is a function that can be used to calculate the Average or Mean of all values of a specified column in the result set. For example, to retrieve the average salary for all employees in the EMPLOYEES table, issue the query: `SELECT AVG(SALARY) FROM EMPLOYEES;`

A good example of grouping would be if for each department ID, we wish to retrieve the number of employees in the department.

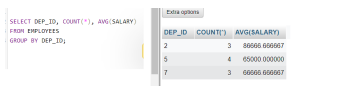
```
SELECT DEP_ID, COUNT(*)  
FROM EMPLOYEES  
GROUP BY DEP_ID;
```



DEP_ID	COUNT(*)
2	3
5	4
7	3

Now, for each department, retrieve the number of employees in the department and the average employee salary in the department. For this, you can use COUNT(*) to retrieve the total count of a column, and AVG() function to compute average salaries, and then GROUP BY.

```
SELECT DEP_ID, COUNT(*), AVG(SALARY)  
FROM EMPLOYEES  
GROUP BY DEP_ID;
```



DEP_ID	COUNT(*)	AVG(SALARY)
2	3	86666.66667
5	4	65000.00000
7	3	66666.66667

You can refine your output by using appropriate labels for the columns of data retrieved. Label the computed columns in the result set of the last problem as NUM_EMPLOYEES and AVG_SALARY.

```
SELECT DEP_ID, COUNT(*) AS "NUM_EMPLOYEES", AVG(SALARY) AS "AVG_SALARY"  
FROM EMPLOYEES  
GROUP BY DEP_ID;
```



DEP_ID	NUM_EMPLOYEES	AVG_SALARY
2	3	86666.66667
5	4	65000.00000
7	3	66666.66667

You can also combine the usage of GROUP BY and ORDER BY statements to sort the output of each group in accordance with a specific parameter. It is important to note that in such a case, ORDER BY clause must be used after the GROUP BY clause. For example, we can sort the result of the previous query by average salary. The SQL query would thus become

```
SELECT DEP_ID, COUNT(*) AS "NUM_EMPLOYEES", AVG(SALARY) AS "AVG_SALARY"  
FROM EMPLOYEES  
GROUP BY DEP_ID  
ORDER BY AVG_SALARY;
```

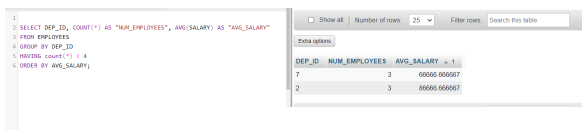
The output of the query should look like:



DEP_ID	NUM_EMPLOYEES	AVG_SALARY
5	4	65000.00000
7	3	66666.66667
2	3	86666.66667

In case you need to filter a grouped response, you have to use the HAVING clause. In the previous example, if we wish to limit the result to departments with fewer than 4 employees, we will have to use HAVING after the GROUP BY, and use the count() function in the HAVING clause instead of the column label.

```
SELECT DEP_ID, COUNT(*) AS "NUM_EMPLOYEES", AVG(SALARY) AS "AVG_SALARY"  
FROM EMPLOYEES  
GROUP BY DEP_ID  
HAVING COUNT(*) < 4  
ORDER BY AVG_SALARY;
```



DEP_ID	NUM_EMPLOYEES	AVG_SALARY
7	3	66666.66667
2	3	86666.66667

Practice Questions

1. Retrieve the list of all employees, first and last names, whose first names start with 'S'.

• [Click here for Solution](#)

```
SELECT F_NAME, L_NAME  
FROM EMPLOYEES  
WHERE F_NAME LIKE 'S%';
```

2. Arrange all the records of the EMPLOYEES table in ascending order of the date of birth.

• [Click here for Solution](#)

```
SELECT *  
FROM EMPLOYEES  
ORDER BY B_DATE;
```

3. Group the records in terms of the department IDs and filter them of ones that have average salary more than or equal to 60000. Display the department ID and the average salary

• [Click here for Solution](#)

```
SELECT DEP_ID, AVG(SALARY)  
FROM EMPLOYEES  
GROUP BY DEP_ID  
HAVING AVG(SALARY) >= 60000;
```

4. For the problem above, sort the results for each group in descending order of average salary.

• [Click here for Solution](#)

```
SELECT DEP_ID, AVG(SALARY)  
FROM EMPLOYEES  
GROUP BY DEP_ID  
HAVING AVG(SALARY) >= 60000  
ORDER BY AVG(SALARY) DESC;
```

Conclusion

Congratulations! You have completed this lab.

By the end of this lab, you are able to:

- Use string patterns for filtering the data retrieved.
- Sort the data retrieved upon one or more parameters using ORDER BY statement.
- Group the data with respect to a parameter.

Author(s)

[Abhishek Lavanya](#)

[Lakshmi Jothi](#)

[Madhu Suresh](#)



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